

URI tests

Business Logic

Server: <http://34.56.164.152:3001>

- IMPORTANT
- Prerequisite: Before testing any URI (except Auth), you must obtain a JWT token by logging in and entering it in Postman under the Authorization tab → Type: Bearer Token.

MODULE: Authentication (Auth)

Test Case-AUTH-01: Successful login as a Dentist.

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Headers	Content-Type: application/json
Body(Json)	{"username": "dentist", "password": "dentist"}
Expected result	HTTP 200
Expected response	{"message": "Login successful.", "token": "<jwt>", "role": "Dentist"}
Validation	The token field exists and the role is "Dentist"

TC-AUTH-02 — Successful login as Administrator

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Body(Json)	{"username": "administrator", "password": "administrator"}
Expected result	HTTP 200
Expected response	{"message": "Login successful.", "token": "<jwt>", "role": "Administrator"}

Validation	The token field exists and the role is "Administrator"
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TC-AUTH-03 — Successful login as Receptionist

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Body(Json)	{"username": "receptionist", "password": "receptionist"}
Expected result	HTTP 200
Expected response	{"message": "Login successful.", "token": "<jwt>", "role": "Receptionist"}
Validation	The token field exists and the role is "Recepcionist"

TC-AUTH-04 — Login with incorrect password

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Body(Json)	{"username": "dentist", "password": "wrongpassword"}
Expected result	HTTP 401
Expected response	{"error": "Authentication failure. Invalid credentials."}

TC-AUTH-05 — Login with no required fields (empty body)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Body(Json)	{}
Expected result	HTTP 400

Expected response	{"error": "username and password are required."}
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TC-AUTH-06 — Login without password

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/login
Body(Json)	{"username": "dentist"}
Expected result	HTTP 400
Expected response	{"error": "username and password are required."}

TC-AUTH-07 — Successful Logout

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/logout
Authorization	Bearer Token → pegar el JWT del login
Expected result	HTTP 200
Expected response	{"message": "Session successfully closed."}
Validation	The same token cannot be used again after logout

TC-AUTH-08 — Logout without token (without Authorization header)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/logout
Authorization	None
Expected result	HTTP 400

Expected response	{"error": "No token found in the Authorization header."}
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TC-AUTH-09 — Register a new valid user

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/register
Body(Json)	{"username": "newdentist", "password": "pass123", "role": "Dentist"}
Expected result	HTTP 201
Expected response	{"success": true, "message": "User registered successfully"}

TC-AUTH-10 — Register with invalid role

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/register
Body(Json)	{"username": "testuser", "password": "pass123", "role": "Nurse"}
Expected result	HTTP 400
Expected response	{"error": "Missing required fields or invalid role."}
Reason	The code only accepts Administrator, Dentist or Receptionist.

TC-AUTH-11 — Register with no required fields

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/auth/register

Body(Json)	{"username": "testuser"}
Expected result	HTTP 400
Expected response	{"error": "Missing required fields or invalid role."}

MODULE: Supplies

Requires login as Administrator and token in Bearer Token.

TC-SUP-01 — Update the expiration status of supplies

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/supplies/status-calculations
Authorization	Bearer Token (Administrator)
Body(Json)	(empty, not required)
Expected result	HTTP 200
Expected response	{"success": true, "message": "Supplies expiration status successfully updated.", "processedItems": <número>}
Validation	success is true, processedItems is a number ≥ 0

TC-SUP-02 — Obtain total inventory valuation

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/asset-valuations
Authorization	Bearer Token (Administrator)
Expected result	HTTP 200
Expected response	{"totalInventoryValue": <número>, "currency": "USD", "calculatedAt": "<ISO date>"}

Validation	Total Inventory Value is a decimal number ≥ 0 , currency is "USD"
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TC-SUP-03 — Obtain replenishment analysis

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/restock-provisions
Authorization	Bearer Token (Administrator)
Expected result	HTTP 200
Expected response	{"consolidatedRestockBudget": <número>, "currency": "USD", "itemsToPurchase": [...]}
Validation	Only items with quantity ≤ 5 appear; unitsToOrder = 15 - actual quantity

TC-SUP-04 — Calculate financial loss due to expired supplies

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/expiration-losses
Authorization	Bearer Token (Administrator)
Expected result	HTTP 200
Expected response	{"totalLossAmount": <número>, "currency": "USD", "expiredItemsCount": <número>, "calculatedAt": "<ISO date>"}
Validation	expiredItemsCount ≥ 0 ; totalLossAmount ≥ 0

TC-SUP-05 — Obtain capital at risk due to upcoming expiration

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/capital-risks
Authorization	Bearer Token (Administrator)
Expected result	HTTP 200
Expected response	{"totalCapitalAtRisk": <número>, "currency": "USD", "riskPeriod": "30 Days or Less", "itemsNearExpirationCount": <número>}
Validation	riskPeriod is always "30 Days or Less"

TC-SUP-06 — Access to Supplies without a token (unauthorized)

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/asset-valuations
Authorization	(none)
Result expected	HTTP 401 or 403
Validation	The server rejects the request without a token.

MODULE: Patients

Requires login as Dentist and token in Bearer Token .

TC-PAT-01 — Calculate pediatric category — School- age child

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category

Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "15/05/2018"}
Result expected	HTTP 200
Expected response	{" calculatedAgeYears ": 8, " calculatedAgeMonths ": 1, " pediatricCategory ": "School-age", " dosageRestrictionFactor ": "0.75x"}
Validation	pediatricCategory = "School- age " for ages between 7-11 years

TC-PAT-02 — Calculate pediatric category — Adult

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "10/01/1990"}
Result expected	HTTP 200
Expected response	{" calculatedAgeYears ": 36, ..., " pediatricCategory ": "Adult", " dosageRestrictionFactor ": "1.0x"}
Validation	pediatricCategory = "Adult" for ≥ 18 years

TC-PAT-03 — Calculate pediatric category — Adolescent

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category
Authorization	Bearer Token (Dentist)

Body (JSON)	{"birthday": "10/03/2010"}
Result expected	HTTP 200
Expected response	{..., " pediatricCategory ": "Adolescent", " dosageRestrictionFactor ": "0.85x"}
Validation	pediatricCategory = " Adolescent " for ages between 12-17 years

TC-PAT-04 — Calculate pediatric category — Toddler

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "10/03/2023"}
Result expected	HTTP 200
Expected response	{..., " pediatricCategory ": "Toddler", " dosageRestrictionFactor ": "0.35x"}
Validation	pediatricCategory = " Toddler " for ages between 1-3 years

TC-PAT-05 — Calculate pediatric category — Invalid date

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": " invalid-date "}
Result expected	HTTP 400

Expected response	<code>{"error": "Valid birthday format required."}</code>
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TC-PAT-06 — Calculate pediatric category — Future date

Field	Value
Method	POST
URL	<code>http://34.56.164.152:3001/fabuladental/patients/pediatric-category</code>
Authorization	Bearer Token (Dentist)
Body (JSON)	<code>{"birthday": "15/05/2030"}</code>
Result expected	HTTP 400
Expected response	<code>{"error": "Valid birthday format required."}</code>
Reason	The code rejects dates after today.

TC-PAT-07 — Validate legal representative — Adult patient (no representative required)

Field	Value
Method	POST
URL	<code>http://34.56.164.152:3001/fabuladental/patients/legal-representative-validation</code>
Authorization	Bearer Token (Dentist)
Body (JSON)	<code>{"birthday": "12/05/2000", "legalRepresentative": ""}</code>
Result expected	HTTP 200
Expected response	<code>{"valid": true, "requiresLegalRepresentative": false, "message": "Legal representative not required."}</code>

TC-PAT-08 — Validate legal representative — Minor without representative (error)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/legal-representative-validation
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "12/05/2012", " legalRepresentative ": " " }
Result expected	HTTP 400
Expected response	{"error": "Patient is underage, legal representative is required."}

TC-PAT-09 — Validate legal representative — Minor WITH representative (successful)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/legal-representative-validation
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "12/05/2012", " legalRepresentative ": "Juan Pérez"}
Result expected	HTTP 200
Expected response	{"valid": true, " requiresLegalRepresentative ": true, "message": "Legal representative provided successfully."}

TC-PAT-10 — Validate legal representative — Invalid date

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/legal-representative-validation

Authorization	Bearer Token (Dentist)
Body (JSON)	{" birthday ": "no-date", " legalRepresentative ": "Juan"}
Result expected	HTTP 400
Expected response	{"error": "Valid birthday format required."}

TC-PAT-11 — Calculate days until birthday — Valid date

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/days-to-birthday
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "15/08/1990"}
Result expected	HTTP 200
Expected response	{" daysUntilBirthday ": < number >, " isBirthdayWeek ": < boolean >}
Validation	daysUntilBirthday ≥ 0 and ≤ 365 ; isBirthdayWeek = true if ≤ 7 days

TC-PAT-12 — Calculate days until birthday — Birthdays this week

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/days-to-birthday
Authorization	Bearer Token (Dentist)
Body (JSON)	{" birthday " : "28/06/1995"} (<i>adjust the date to today's date $\pm$ 3 days</i>)
Result expected	HTTP 200

Expected response	<code>{" daysUntilBirthday ": <0-7>, " isBirthdayWeek ": true }</code>
Validation	isBirthdayWeek must be true

TC-PAT-13 — Calculate days until birthday — Invalid format

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/days-to-birthday
Authorization	Bearer Token (Dentist)
Body (JSON)	<code>{"birthday": "1990-08-15-XX"}</code>
Result expected	HTTP 400
Expected response	<code>{"error": "Valid birthday format required."}</code>

TC-PAT-14 — Calculate senior discount — Eligible patient (≥ 65 years)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/senior-discount
Authorization	Bearer Token (Dentist)
Body (JSON)	<code>{"birthday": "10/01/1954"}</code>
Result expected	HTTP 200
Expected response	<code>{" suggestedDiscountFactor ": 0.5, " isEligibleForDiscount ": true }</code>
Validation	isEligibleForDiscount = true and suggestedDiscountFactor = 0.5

TC-PAT-15 — Calculate senior discount — Patient NOT eligible (< 65 years)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/senior-discount
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "10/01/1990"}
Result expected	HTTP 200
Expected response	{" suggestedDiscountFactor ": 0, " isEligibleForDiscount ": false}
Validation	isEligibleForDiscount = false and suggestedDiscountFactor = 0

TC-PAT-16 — Calculate senior discount — Invalid date

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/senior-discount
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday ": " "}
Result expected	HTTP 400
Expected response	{"error": "Valid birthday format required."}

TC-PAT-17 — Estimate consultation time — Very young patient (< 5 years) with long ratio

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/consultation-time-estimation

Authorization	Bearer Token (Dentist)
Body (JSON)	{" birthday ": "10/01/2021", " reasonForConsultation ": "The patient presents with acute pain in the lower right molar that prevents him from sleeping and has severe inflammation in the gums."}
Result expected	HTTP 200
Expected response	{" estimatedConsultationMinutes ": 35, " baseMinutes ": 15}
Validation	Base=15 + 10 (less than 5 years old) + 10 (ratio > 50 chars) = 35 min

TC-PAT-18 — Estimate consultation time — Adult with short ratio

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/consultation-time-estimation
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "01/10/1985", " reasonForConsultation ": " Teeth cleaning "}
Result expected	HTTP 200
Expected response	{" estimatedConsultationMinutes ": 15, " baseMinutes ": 15}
Validation	Without modifiers, just the base = 15 min

TC-PAT-19 — Estimate consultation time — Older adult (> 70 years) with short ratio

Field	Value
Method	POST

URL	http://34.56.164.152:3001/fabuladental/patients/consultation-time-estimation
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "10/01/1950", "reasonForConsultation": "Review"}
Result expected	HTTP 200
Expected response	{"estimatedConsultationMinutes": 25, "baseMinutes": 15}
Validation	Base=15 + 10 (over 70 years old) = 25 min

TC-PAT-20 — Estimate consultation time — Invalid date

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/consultation-time-estimation
Authorization	Bearer Token (Dentist)
Body (JSON)	{"birthday": "invalid-date", "reasonForConsultation": "Pain"}
Result expected	HTTP 400
Expected response	{"error": "Valid birthday format required."}

TC-PAT-21 — Calculate contact priority — High priority (phone + urgency)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/contact-priority

Authorization	Bearer Token (Dentist)
Body (JSON)	{" phone ": "0991234567", " reasonForConsultation ": "I have a lot of pain and bleeding, please, it's an emergency."}
Result expected	HTTP 200
Expected response	{" contactPriorityScore ": 60, " requiresImmediateAttention ": true}
Validation	10 (phone $\geq$ 10 digits) + 50 (urgent words) = 60 points

TC-PAT-22 — Calculate contact priority — Low priority (no phone, no urgency)

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/contact-priority
Authorization	Bearer Token (Dentist)
Body (JSON)	{" phone ": "", " reasonForConsultation ": "I want a dental cleaning."}
Result expected	HTTP 200
Expected response	{" contactPriorityScore ": 0, " requiresImmediateAttention ": false}
Validation	With no points assigned, requiresImmediateAttention = false

TC-PAT-23 — Calculate contact priority — Phone only, no urgency

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/contact-priority
Authorization	Bearer Token (Dentist)

Body (JSON)	<code>{"phone": "0991234567", " reasonForConsultation ": "Routine check-up . "}</code>
Result expected	HTTP 200
Expected response	<code>{" contactPriorityScore ": 10, " requiresImmediateAttention ": false}</code>
Validation	Only 10 points per phone; requiresImmediateAttention = false (requires ≥ 50)

TC-PAT-24 — Calculate contact priority — Urgent word only, no phone

Field	Value
Method	POST
URL	<code>http://34.56.164.152:3001/fabuladental/patients/contact-priority</code>
Authorization	Bearer Token (Dentist)
Body (JSON)	<code>{" phone ": "", " reasonForConsultation ": "It's an emergency, heavy bleeding."}</code>
Result expected	HTTP 200
Expected response	<code>{" contactPriorityScore ": 50, " requiresImmediateAttention ": true}</code>
Validation	Only 50 points for urgent words; requiresImmediateAttention = true

MODULE: Payments

Requires login as Receptionist and token in Bearer Token.

TC-PAY-01 — Obtain payment history

Field	Value
Method	GET

URL	http://34.56.164.152:3001/fabuladental/payments
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 200
Expected response	JSON array with objects { paymentID , patientID , amount, status }
Validation	status is one of: " Completed ", " Partial " or " Pending "

TC-PAY-02 — Get Payment by ID — Existing ID

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/5
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 200
Expected response	{"success": true, "data ": { "id": "5", " patientId ": "...", "amount": 50, "type": "Final", "status": "Completed", "date": "..."} }
Validation	success = true; data.id = "5"

TC-PAY-03 — Get payment by ID — ID does not exist

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/99999
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 404
Expected response	{"error": "Payment not found."}

TC-PAY-04 — Obtain payments per patient — Valid patient ID (existing ID card)

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/patients/1727417859
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 200
Expected response	{"success": true, "data" : [{ "id": "...", "amount": ..., "type": "...", "status": "...", "date": " ..." }]}
Validation	success = true; data is an array with at least 1 element

TC-PAY-05 — Obtain payments per patient — Patient without payments / does not exist

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/patients/000000000
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 404
Expected response	{"error": "Patient not found or no payment records discovered."}

TC-PAY-06 — Get payments per patient — Invalid ID (does not have 10 digits)

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/patients/123
Authorization	Bearer Token (Receptionist)

Result expected	HTTP 400
Expected response	{"error": "Invalid Patient ID parameter."}
Reason	The code validates that the ID is exactly 10 numeric digits.

TC-PAY-07 — Obtain payments per patient — ID with letters (invalid)

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments/patients/abc defghij
Authorization	Bearer Token (Receptionist)
Result expected	HTTP 400
Expected response	{"error": "Invalid Patient ID parameter."}

ACCESS CONTROL TESTS (Cross-Role)

TC-ROLE-01 — Dentist attempting to access Supplies

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/supplies/asset-valuation s
Authorization	Bearer Token (Dentist)
Result expected	HTTP 403
Validation	The system blocks access to modules outside the assigned role.

TC-ROLE-02 — Receptionist trying to access Patients

Field	Value
Method	POST
URL	http://34.56.164.152:3001/fabuladental/patients/pediatric-category
Authorization	Bearer Token (Receptionist)
Body (JSON)	{"birthday": "10/01/1990"}
Result expected	HTTP 403
Validation	The system blocks access to modules outside the assigned role.

TC-ROLE-03 — Administrator attempting to access Payments

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments
Authorization	Bearer Token (Administrator)
Result expected	HTTP 403
Validation	The system blocks access to modules outside the assigned role.

TC-ROLE-04 — Invalid/expired token on any protected URI

Field	Value
Method	GET
URL	http://34.56.164.152:3001/fabuladental/payments
Authorization	Bearer Token → eyJININVALIDTOKEN.fake.signature
Result expected	HTTP 401 or 403
Validation	The server rejects invalid tokens

CRUD Operations

<http://136.113.240.33:3000>

Auth header required on every request: **x-api-key:admin**

Color Legend

2xx — Success	4xx — Client Error	5xx — Server Error	POST method	GET method	PUT method	DELETE method
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1. Users

The Users endpoints are consumed internally by the Business Logic server to validate credentials and register system users. The API key x-api-key: admin must be sent in the Headers tab.

1.1 POST /fabuladental/users — Register a New User

POST	/fabuladental/users	<i>Registers a new system user with an encrypted password and an assigned business role (Administrator, Dentist, or Receptionist).</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	USR-REG-01	Register with all valid fields and a correct bcrypt-hashed password	{"username": "dentist", "password": "\$2b\$10\$4Aq..ze", "role": "Dentist"}	201	{"success": true, "message": "User registered successfully"}
2	USR-REG-02	Register with role: Administrator	{"username": "admin2", "password": "\$2b\$10\$...", "role": "Administrator"}	201	{"success": true, "message": "User registered successfully"}
3	USR-REG-03	Register with role: Receptionist	{"username": "recep1", "password": "\$2b\$10\$...", "role": "Receptionist"}	201	{"success": true, "message": "User registered successfully"}

4	USR-REG-04	Missing username field	{"password":"\$2b\$10\$...","role":"Dentist"}	400	{"error":"Missing required fields or invalid role."}
5	USR-REG-05	Missing password field	{"username":"dentist","role":"Dentist"}	400	{"error":"Missing required fields or invalid role."}
6	USR-REG-06	Invalid role value (not one of the 3 accepted roles)	{"username":"test","password":"\$2b\$10\$...","role":"SuperAdmin"}	400	{"error":"Missing required fields or invalid role."}
7	USR-REG-07	Empty request body (no fields sent)	{}	400	{"error":"Missing required fields or invalid role."}
8	USR-REG-08	Database / server error during insertion	(simulate DB failure)	500	{"error":"Database error while registering user."}

1.2 GET /fabuladental/users/:username — Get User by Username

GET	/fabuladental/users/:username	Retrieves the full information of a single user identified by their unique username to support the authentication validation process.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	USR-GET-01	Fetch an existing user by username	/fabuladental/users/dentist	200	{"id":11,"username":"dentist","role":"Dentist",...}
2	USR-GET-02	Fetch Administrator user	/fabuladental/users/administrator	200	{"id":...,"username":"administrator","role":"Administrator",...}
3	USR-GET-03	Fetch Receptionist user	/fabuladental/users/receptionist	200	{"id":...,"username":"receptionist","role":"Receptionist",...}

4	USR-GET-04	Username that does not exist in the database	/fabuladental/users/ghost_user	404	{"error": "User not found."}
5	USR-GET-05	Database / server error during fetch	(simulate DB failure)	500	{"error": "Database error while fetching user."}

2. Supplies

2.1 GET /fabuladental/supplies — Get All Supplies

GET	/fabuladental/supplies	<i>Retrieves the complete list of dental supplies directly from the database.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	SUP-GET-01	Retrieve full inventory list (happy path)	GET /fabuladental/supplies	200	Array of all supply objects
2	SUP-GET-02	Server / database error during fetch	(simulate DB failure)	500	{"error": "Unable to fetch inventory."}

2.2 GET /fabuladental/supplies/quantity-thresholds/:maxQuantity — Filter by Quantity

GET	/fabuladental/supplies/quantity-thresholds/:maxQuantity	<i>Returns supplies whose current stock is less than or equal to the maxQuantity value passed in the URL.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
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1	SUP-QTY-01	Filter with maxQuantity=5 — returns items with qty ≤ 5	/fabuladental/supplies/quantity-thresholds/5	200	Array of matching supply objects
2	SUP-QTY-02	Filter with maxQuantity=2	/fabuladental/supplies/quantity-thresholds/2	200	Array of supplies with quantity ≤ 2
3	SUP-QTY-03	Filter with maxQuantity=0 — no items at zero stock	/fabuladental/supplies/quantity-thresholds/0	200	[] (empty array)
4	SUP-QTY-04	Non-numeric maxQuantity parameter	/fabuladental/supplies/quantity-thresholds/abc	400	{"error": "Invalid maximum quantity format."}
5	SUP-QTY-05	Negative value as maxQuantity	/fabuladental/supplies/quantity-thresholds/-1	400	{"error": "Invalid maximum quantity format."}
6	SUP-QTY-06	Database error during filter query	(simulate DB failure)	500	{"error": "Database error while filtering supply quantities."}

2.3 GET /fabuladental/supplies/statuses/:statusValue — Filter by Status

GET	/fabuladental/supplies/statuses/:statusValue	Retrieves records matching the exact status string: Expired, NextExpiration, or Current.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	SUP-STA-01	Filter by status: Expired	/fabuladental/supplies/statuses/Expired	200	Array of expired supply objects
2	SUP-STA-02	Filter by status: NextExpiration	/fabuladental/supplies/statuses/NextExpiration	200	Array of near-expiry supplies
3	SUP-STA-03	Filter by status: Current	/fabuladental/supplies/statuses/Current	200	Array of current/active supplies

4	SUP-STA-04	Non-existent status value	/fabuladental/supplies/statuses/Vencido	400	{"error": "Provided status value does not exist."}
5	SUP-STA-05	Empty status string	/fabuladental/supplies/statuses/	400	{"error": "Provided status value does not exist."}
6	SUP-STA-06	Database error during filter	(simulate DB failure)	500	{"error": "Database error while filtering supply statuses."}

2.4 POST /fabuladental/supply — Register a Dental Supply

POST	/fabuladental/supply	<i>Registers a new dental supply into the inventory with name, quantity, unit cost, order date, and expiration date.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	SUP-ADD-01	Register a supply with all valid fields	{"supplyName": "Resin", "quantity": 10, "unitCost": 5.50, "orderDate": "2026-06-09", "expirationDate": "2027-06-09"}	201	{"success": true, "message": "Supply added"}
2	SUP-ADD-02	Register with minimum quantity (1)	{"supplyName": "Gloves", "quantity": 1, "unitCost": 0.10, "orderDate": "2026-01-01", "expirationDate": "2027-01-01"}	201	{"success": true, "message": "Supply added"}
3	SUP-ADD-03	Missing supplyName field	{"quantity": 10, "unitCost": 5.50, "orderDate": "2026-06-09", "expirationDate": "2027-06-09"}	400	{"error": "Missing supply data."}

4	SUP-ADD-04	Missing quantity field	<code>{"supplyName": "Resin", "unitCost": 5.50, "orderDate": "2026-06-09", "expirationDate": "2027-06-09"}</code>	400	<code>{"error": "Missing supply data."}</code>
5	SUP-ADD-05	Missing expirationDate field	<code>{"supplyName": "Resin", "quantity": 10, "unitCost": 5.50, "orderDate": "2026-06-09"}</code>	400	<code>{"error": "Missing supply data."}</code>
6	SUP-ADD-06	Empty request body	<code>{}</code>	400	<code>{"error": "Missing supply data."}</code>
7	SUP-ADD-07	Database error during insertion	(simulate DB failure)	500	<code>{"error": "Could not add supply."}</code>

2.5 PUT /fabuladental/supplies/:id — Update a Supply

PUT	/fabuladental/supplies/:id	<i>Edits an existing supply in the inventory by updating any of its modifiable attributes.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	SUP-UPD-01	Update all fields of an existing supply	<code>PUT /supplies/7 {"supplyName": "Resin Z350", "quantity": 15, "unitCost": 6.25, "orderDate": "2026-06-09", "expirationDate": "2027-06-09"}</code>	200	<code>{"success": true, "message": "Supply updated"}</code>
2	SUP-UPD-02	Update only the quantity of an existing supply	<code>PUT /supplies/7 {"quantity": 20}</code>	200	<code>{"success": true, "message": "Supply updated"}</code>
3	SUP-UPD-03	Non-numeric ID in URL path	<code>PUT /fabuladental/supplies/abc</code>	400	<code>{"error": "Invalid supply ID or quantity."}</code>
4	SUP-UPD-04	Supply ID that does not exist	<code>PUT /fabuladental/supplies/99999</code>	404	<code>{"error": "Supply not found."}</code>

2.6 DELETE /fabuladental/supplies/:id — Delete a Supply

DELETE	/fabuladental/supplies/:id	Removes a dental supply from the inventory using its unique numeric ID.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	SUP-DEL-01	Delete an existing supply by valid ID	DELETE /fabuladental/supplies/7	200	{"success":true,"message":"Supply deleted"}
2	SUP-DEL-02	No ID provided in the URL path	DELETE /fabuladental/supplies/	400	{"error":"Supply ID required."}
3	SUP-DEL-03	Supply ID that does not exist in the database	DELETE /fabuladental/supplies/99999	404	{"error":"Supply not found."}
4	SUP-DEL-04	Database / server error during deletion	(simulate DB failure)	500	{"error":"Deletion failed."}

3. Payments

3.1 PUT /fabuladental/payments — Get All Payments

PUT	/fabuladental/payments	⚠ Uses PUT method. Retrieves the complete raw list of all payment records from the database.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAY-GET-01	Retrieve all payment records (happy path) — uses PUT method	PUT /fabuladental/payments (no body)	200	Array of all payment objects
2	PAY-GET-02	Database / server error during fetch	(simulate DB failure)	500	{"error":"Unable to fetch payment history."}

3.2 POST /fabuladental/payments — Record a Payment

POST	/fabuladental/payments	<i>Records a new patient payment (Deposit or Final) with amount, payment method, and status.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAY-REC-01	Record a Final payment via Cash — all valid fields	<code>{"patientID": "1727417851", "amount": "50", "date": "2026-05-13", "paymentType": "Final", "paymentMethod": "Cash", "status": "Completed"}</code>	201	<code>{"success": true, "message": "Payment recorded"}</code>
2	PAY-REC-02	Record a Deposit payment via Transfer	<code>{"patientID": "1727417851", "amount": "25", "date": "2026-06-01", "paymentType": "Deposit", "paymentMethod": "Transfer", "status": "Partial"}</code>	201	<code>{"success": true, "message": "Payment recorded"}</code>
3	PAY-REC-03	Missing patientID field	<code>{"amount": "50", "date": "2026-05-13", "paymentType": "Final", "paymentMethod": "Cash", "status": "Completed"}</code>	400	<code>{"error": "Invalid payment data."}</code>
4	PAY-REC-04	Missing amount field	<code>{"patientID": "1727417851", "date": "2026-05-13", "paymentType": "Final", "paymentMethod": "Cash", "status": "Completed"}</code>	400	<code>{"error": "Invalid payment data."}</code>
5	PAY-REC-05	Empty request body	<code>{}</code>	400	<code>{"error": "Invalid payment data."}</code>
6	PAY-REC-06	Database / server error during insertion	(simulate DB failure)	500	<code>{"error": "Could not record payment."}</code>

3.3 PUT /fabuladental/payments/:paymentId — Update a Payment

PUT	<code>/fabuladental/payments/:paymentId</code>	<i>Updates a recorded payment to correct errors or update the status of a specific payment record.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAY-UPD-01	Update all fields of an existing payment record	PUT /payments/7 { "patientID": "1725629313", "amount": "45", "date": "2020-10-01T00:00:00.000Z", "paymentType": "Deposit", "paymentMethod": "Transfer", "status": "Partial" }	200	{ "success": true, "message": "Payment updated" }
2	PAY-UPD-02	Update only the status of a payment	PUT /payments/7 { "status": "Completed" }	200	{ "success": true, "message": "Payment updated" }
3	PAY-UPD-03	Non-numeric payment ID in URL	PUT /fabuladental/payments/abc	400	{ "error": "Invalid payment ID or data." }
4	PAY-UPD-04	Payment ID that does not exist	PUT /fabuladental/payments/9999	404	{ "error": "Payment record not found." }

3.4 DELETE /fabuladental/payments — Delete a Payment

DELETE	<code>/fabuladental/payments</code>	<i>Deletes a payment record using its id sent in the request body as JSON.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAY-DEL-01	Delete an existing payment record by valid ID	DELETE /fabuladental/payments { "id": 2 }	200	{ "success": true, "message": "Payment deleted" }
2	PAY-DEL-02	Body sent but id field is missing	DELETE /fabuladental/payments { }	400	{ "error": "Payment ID required." }

3	PAY-DEL-03	Payment ID that does not exist in the database	DELETE /fabuladental/payments {"id":99999}	404	{"error":"Payment not found."}
4	PAY-DEL-04	Database / server error during deletion	(simulate DB failure)	500	{"error":"Deletion failed."}

4. Patients

4.1 POST /fabuladental/patients — Register a New Patient

POST	/fabuladental/patients	<i>Registers a new patient with their DNI, full name, birthday, contact info, gender, and reason for consultation.</i>
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAT-REG-01	Register a patient with all required fields	{"patientID":"1727095416","fullName":"Victoria Diaz","phone":"0984284522","email":"cv@espe.edu.ec","birthday":"20/03/2003","gender":"Female","reasonForConsultation":"gum pain"}	201	{"success":true,"message":"Patient registered successfully"}
2	PAT-REG-02	Register a male patient	{"patientID":"1727095417","fullName":"Carlos Ruiz","phone":"0991234567","email":"c@test.com","birthday":"15/08/1990","gender":"Male","reasonForConsultation":"toothache"}	201	{"success":true,"message":"Patient registered successfully"}
3	PAT-REG-03	Missing fullName field	{"patientID":"1727095418","phone":"0984284522","birthday":"20/03/2003","gender":"Female","reasonForConsultation":"gum pain"}	400	{"error":"Missing required fields."}
4	PAT-REG-04	Missing patientID field	{"fullName":"Victoria Diaz","phone":"0984284522","birthday":"20/03/2003","gender":"Female","reasonForConsultation":"gum pain"}	400	{"error":"Missing required fields."}

5	PAT-REG-05	Missing birthday field	{"patientID":"1727095419","fullName":"Victoria Diaz","phone":"0984284522","gender":"Female","reasonForConsultation":"gum pain"}	400	{"error":"Missing required fields."}
6	PAT-REG-06	Empty request body	{}	400	{"error":"Missing required fields."}
7	PAT-REG-07	Database / server error during insertion	(simulate DB failure)	500	{"error":"Something went wrong. Please try again later."}

4.2 GET /fabuladental/patients — Get All Patients

GET	/fabuladental/patients	Retrieves the list of all registered patients to populate the dentist's patient table.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAT-GET-01	Retrieve all registered patients (happy path)	GET /fabuladental/patients	200	Array of all patient objects
2	PAT-GET-02	Server / database error during fetch	(simulate DB failure)	400	{"error":"Unable to fetch patients."}

4.3 GET /fabuladental/patients/:patientID — Get Patient by ID

GET	/fabuladental/patients/:patientID	Retrieves the full information of a single patient identified by their DNI number.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
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1	PAT-GID-01	Fetch an existing patient by valid DNI	GET /fabuladental/patients/1727095414	200	{"patientID":"1727095414","fullName":"Victoria Diaz",...}
2	PAT-GID-02	DNI that does not exist in the database	GET /fabuladental/patients/0000000000	404	{"error":"Patient not found."}
3	PAT-GID-03	Server error during patient lookup	(simulate server failure)	505	{"error":"Unable to fetch patient."}

4.4 PUT /fabuladental/patients/:patientId — Update Patient Information

PUT	/fabuladental/patients/:patientId	Updates an existing patient's information using their DNI as the URL parameter.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAT-UPD-01	Update all fields of an existing patient	PUT /patients/1727095414 {"patientID":"1727095414","fullName":"Victoria Diaz","phone":"0984284522","email":"cv@espe.edu.ec","birthday":"20/03/2003","gender":"Female","reasonForConsultation":"gum pain"}	200	{"success":true,"message":"Patient updated"}
2	PAT-UPD-02	Update only the phone number	PUT /patients/1727095414 {"phone":"0991111111"}	200	{"success":true,"message":"Patient updated"}
3	PAT-UPD-03	No patient ID provided in the URL	PUT /fabuladental/patients/	400	{"error":"Patient ID is required."}
4	PAT-UPD-04	Patient DNI that does not exist	PUT /fabuladental/patients/9999999999	404	{"error":"Patient"}

					not found."}
5	PAT-UPD-05	Database / server error during update	(simulate DB failure)	500	{"error": "Something went wrong."}

4.5 DELETE /fabuladental/patients/:patientId — Delete a Patient

DELETE	/fabuladental/patients/:patientId	Deletes a patient record using their DNI number in the URL path.
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#	Test Case ID	Description	Input / Payload	Expected Status	Expected Response
1	PAT-DEL-01	Delete an existing patient by valid DNI	DELETE /fabuladental/patients/1727095419	200	{"success":true,"message":"Patient deleted"}
2	PAT-DEL-02	No patient ID in the URL path	DELETE /fabuladental/patients/	400	{"error":"Invalid or missing patient ID."}
3	PAT-DEL-03	Patient DNI that does not exist	DELETE /fabuladental/patients/999999999	404	{"error":"Patient not found."}
4	PAT-DEL-04	Database / server error during deletion	(simulate DB failure)	500	{"error":"Deletion failed."}